

WILL SMITH

willsmith.nz portfolio — live demos, source, commit history

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SUMMARY

Software engineer with six months of full-time commercial experience as an IT contractor at an MSP — shipping production software, operating live SaaS infrastructure, and designing a self-hosted hosting platform that replaced third-party hosting. Backed by a Computer Science / Economics conjoint degree from the University of Auckland, and an independently-built portfolio spanning embedded systems, full-stack web apps, and applied ML — most of it designed, built, and operated solo outside of work. Comfortable across the stack — from vSphere and Terraform infrastructure to React front ends to fixed-point arithmetic on bare-metal hardware.

WORK EXPERIENCE

IT Consultant (Contractor) — Novatec Solutions, Auckland

February 2026 – Present

An MSP for small-to-medium businesses. Primary technical contractor, full-time, working alongside the owner across software development and infrastructure.

Software development

- Built an invoicing automation system integrating Pax8 (cloud billing) and Xero (accounting) via Windmill — fuzzy-matches customers across systems and drafts invoices with LLM-assisted content cleanup; live, with final polish ongoing.
- Built a Python HTTP server exposing Microsoft 365 (mail, calendar, OneDrive, tasks) via per-user OAuth — deployed and in active use.
- Built a FastAPI email backend serving contact-form submissions across multiple client sites.
- Own the infrastructure for a live, multi-tenant SaaS product — Docker/Terraform VM provisioning, secrets management, and identity/auth design.
- Operate that infrastructure in production — diagnosing and fixing live incidents, building its observability layer (structured audit logging + Prometheus monitoring), and running a security-incident response (credential rotation and git-history remediation after a leaked key).
- Built company and client web properties in TypeScript/React, collaborating with another developer on the company's own marketing site.

Infrastructure & hosting

- Designed and stood up a self-hosted client-hosting platform (Coolify on on-prem, UPS-backed ESXi virtual machines), migrating client sites off third-party hosting and eliminating its recurring cost.
- Built **novatec.co.nz** end to end — site, DNS configuration, and Cloudflare Tunnels.
- Migrated several existing client WordPress sites onto the new platform.
- Hands-on VM administration (ESXi/vSphere) and core networking (WAPs, switches, routers, DHCP).

PROJECTS

Live demos, source, and commit activity: **willsmith.nz**

PromptTech AI Sandbox — Capstone Project (Industry Client), 2025 — **graded A**

- Dockerised full-stack web application for comparing prompts and parameters across generative AI models.
- Integrated AWS Bedrock and Cognito (plus Google OAuth), deployed on AWS EC2; REST API in Node.js/Express with a React front end.
- Industry client: PromptTech (reference below).

Recipe Web App — live at willscookbook.nz

github.com/QuickWaller/recipe-website-v2

- Full rewrite of an earlier recipe app: Astro + React islands, Drizzle ORM/Postgres, Docker Compose. Unit, end-to-end, and visual-regression tests, gated by CI.
- Self-hosted on a personal Proxmox home lab (VLAN-segmented, Tailscale-only admin access), deployed via Coolify and Cloudflare Tunnel — the same hosting stack built professionally at Novatec, run independently at home.

Hiking Data Logger / Weather Pod

github.com/QuickWaller/hiking-weather-pod

- Low-power field logger: GPS, barometric/weather sensing, e-ink display, rule-based and ML weather forecasting, SD-card sync.
- Production-scale ML forecasting pipeline: LightGBM trained on ERA5-Land reanalysis against GPM satellite rainfall, ~31M training rows — deployed to the pod's firmware, not lab-only.

- Evaluated as real research, not a demo: skill scored against a climatology baseline (Brier/CRPS), sensor-noise simulation, and model bake-offs to choose the production configuration, with feature parity to the on-device C++ firmware via cffi.

ESP32 FM Synthesiser

github.com/QuickWaller/esp32-fm-synth · [.../esp32-synth-fixed-point-wip](https://github.com/QuickWaller/esp32-synth-fixed-point-wip)

- Real-time FM audio synthesis on ESP32, streamed over Bluetooth A2DP — table-lookup oscillator, ADSR envelope, multi-voice mixing.
- Separate fixed-point rework: a from-scratch Q16.16/Q32.0 fixed-point ADSR envelope generator, working around the absence of hardware floating point.

ESP32-CAM Face Detection

- Low-cost microcontroller streaming video over WebSocket to a FastAPI/YOLOv8 backend for real-time face detection.
- Backpressure-aware streaming design (bounded queue, drop-oldest-frame) to handle network variability.

Vintage Story Game Server

github.com/QuickWaller/vintage-story-server

- Enhanced fork adding automated mod management (40+ mods), Coolify deployment, and structured logging.

TECHNICAL SKILLS

Languages	C, C++, C#, Python, Java, TypeScript/JavaScript, HTML/CSS
Frameworks	React, Node.js, Express, FastAPI, Flask, .NET, Arduino framework/PlatformIO, Godot, Unity
Cloud & infrastructure	AWS (EC2, Cognito, Bedrock), Docker, Terraform, Coolify, VMware ESXi/vSphere, Proxmox VE, Cloudflare Tunnels, Tailscale, Linux administration, Git, core networking (VLANs, DHCP, routing)
Data & APIs	PostgreSQL, REST, WebSockets, OAuth, Stripe
Embedded	ESP32/ESP32-CAM, fixed-point arithmetic, real-time audio DSP, low-power design, sensor integration
AI/ML	Generative AI API integration, applied ML forecasting pipelines (LightGBM, satellite/reanalysis data fusion, skill-score evaluation), embedded computer vision, machine learning fundamentals

EDUCATION

University of Auckland — Bachelor of Commerce / Bachelor of Science (Conjoint), Computer Science & Economics
2021 – 2026, graduated

- **COMPSCI 399 Capstone: Computer Science** — A (industry-client project for PromptTech, see Projects)
- **COMPSCI 361 Machine Learning** — A
- **COMPSCI 235 Software Development Methodologies** — A-

Kristin School — NCEA Levels 1–3 Graduated 2020

- Dux 2019 and 2020 · Kristin Cup for Excellence in Mathematics, 2020 · Achieved with Excellence at all levels.

ADDITIONAL EXPERIENCE

Waiter / Line Cook — Long Bay Surf Club November 2025 – February 2026

Volunteer, Breakfast Service — Auckland City Mission January 2025 – August 2025

- Supported Auckland's homeless and vulnerable community at a soup-kitchen-style breakfast service — a high-stress environment requiring reliability and composure.

Calculus & Statistics Tutor — Self-managed February 2024 – July 2025

- Ran a one-on-one NCEA tutoring business — found students, scheduled and priced sessions, wrote lesson plans and materials.

Commis Chef — Beau 2022 – 2024

- High-end French cuisine; trained new team members; managed ordering and stock.

Sales Assistant — Michael Hill Jewellers 2019 – 2021

Salesperson & Delivery Driver — Dough Boys Bakery 2019 – 2021

- Self-managed logistics and sales at markets.

Team Member — KFC 2018 – 2020

REFERENCES

Amir Mohammadi — AI Adoption and Development Lead, PromptTech — amir@prompttech.ai

- Worked with Amir on the PromptTech AI Sandbox capstone project; happy to speak to technical and soft skills.